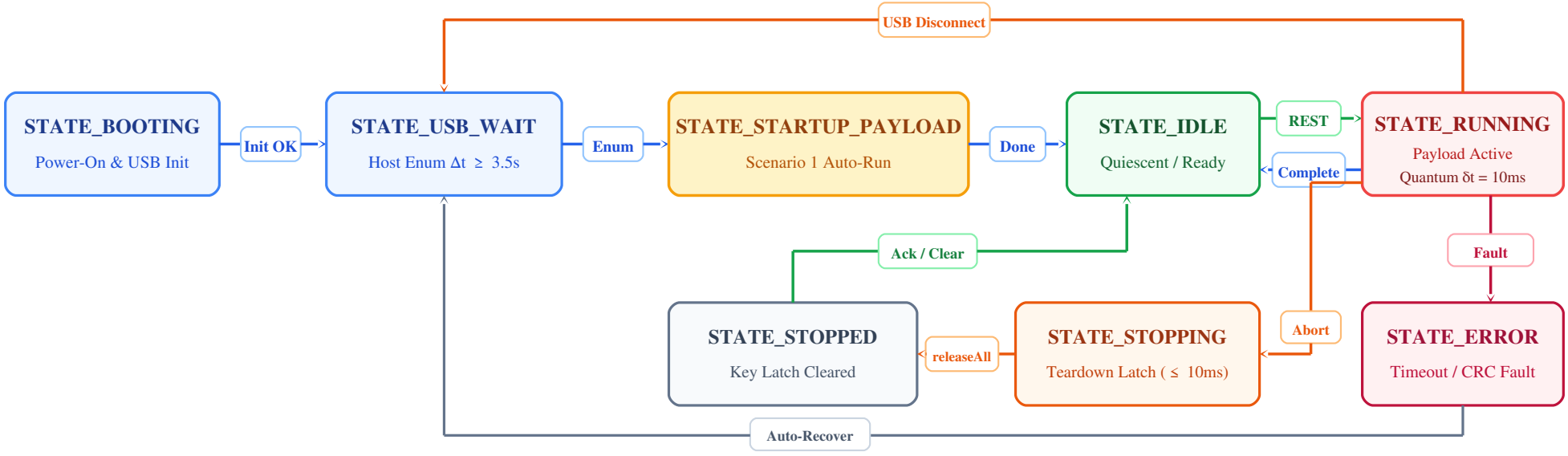

USB Disconnect
STATE_BOOTING
Power-On & USB Init
Init OK
STATE_USB_WAIT
Host Enum Δt ≥ 3.5s
Enum
STATE_STARTUP_PAYLOAD
Scenario 1 Auto-Run
Done
STATE_IDLE
Quiescent / Ready
REST
Complete
STATE_RUNNING
Payload Active
Quantum δt = 10ms
Fault
Ack / Clear
STATE_STOPPED
Key Latch Cleared
releaseAll
STATE_STOPPING
Teardown Latch (≤ 10ms)
Abort
STATE_ERROR
Timeout / CRC Fault
Auto-Recover